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CODEARCHAEOLOGIST

AST-Guided, LLM-Powered Legacy Code Comprehension & Automated Documentation

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Guide

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Problem Area

Developers spend the majority of their time reading and understanding existing code rather than writing new code.
Legacy systems are often poorly documented, and the developers who originally wrote them are frequently unavailable.
Manually tracing program logic across large codebases is slow, error-prone, and delays new-developer onboarding.
Existing tools describe code only structurally or numerically — they cannot explain, in plain language, what the code does or why.
There is a need for a tool that automatically explains code and generates documentation grounded in the program's real structure.

Project Statement

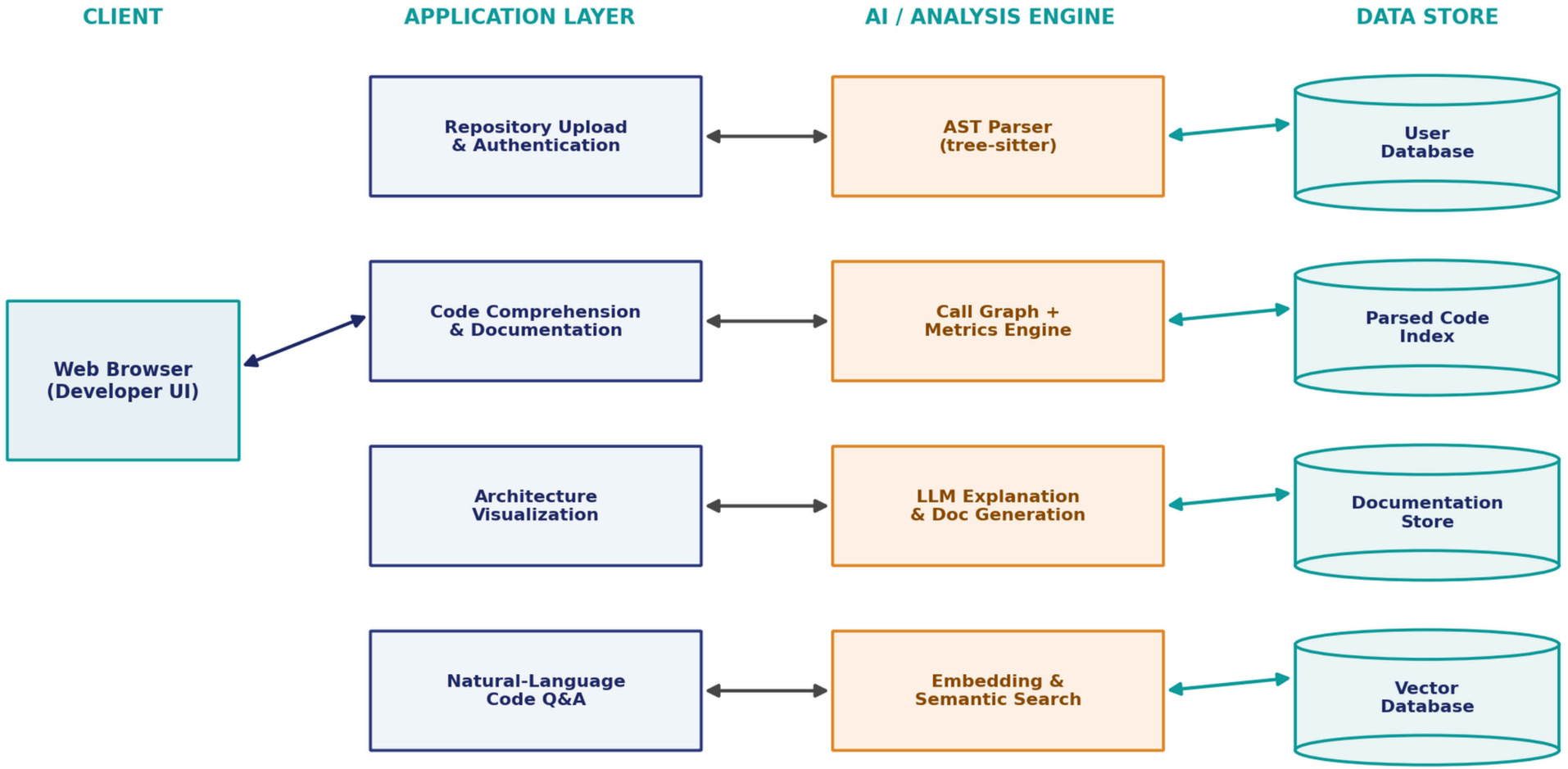
- Modern software development rarely begins from scratch; developers spend most of their time reading, understanding, and modifying code that already exists. In large or ageing systems this code is frequently undocumented, and the engineers who originally wrote it may no longer be available. As a result, developers must manually trace logic across many files to answer even simple questions, such as what a module does or how components depend on one another. This comprehension effort is a well-known bottleneck that slows onboarding, delays feature development, and increases the likelihood of introducing defects. Traditional static-analysis tools can extract accurate structural information — abstract syntax trees, call graphs, and complexity metrics — but they cannot explain code in natural language, which is what developers most often need. Large Language Models can summarise code fluently, yet used alone they may hallucinate and struggle with large codebases. CodeArchaeologist addresses this gap by combining reliable static analysis with LLM-based explanation: it parses a codebase, builds a structured representation, and uses that grounded context to generate trustworthy explanations, documentation, architecture diagrams, and natural-language answers about the code.

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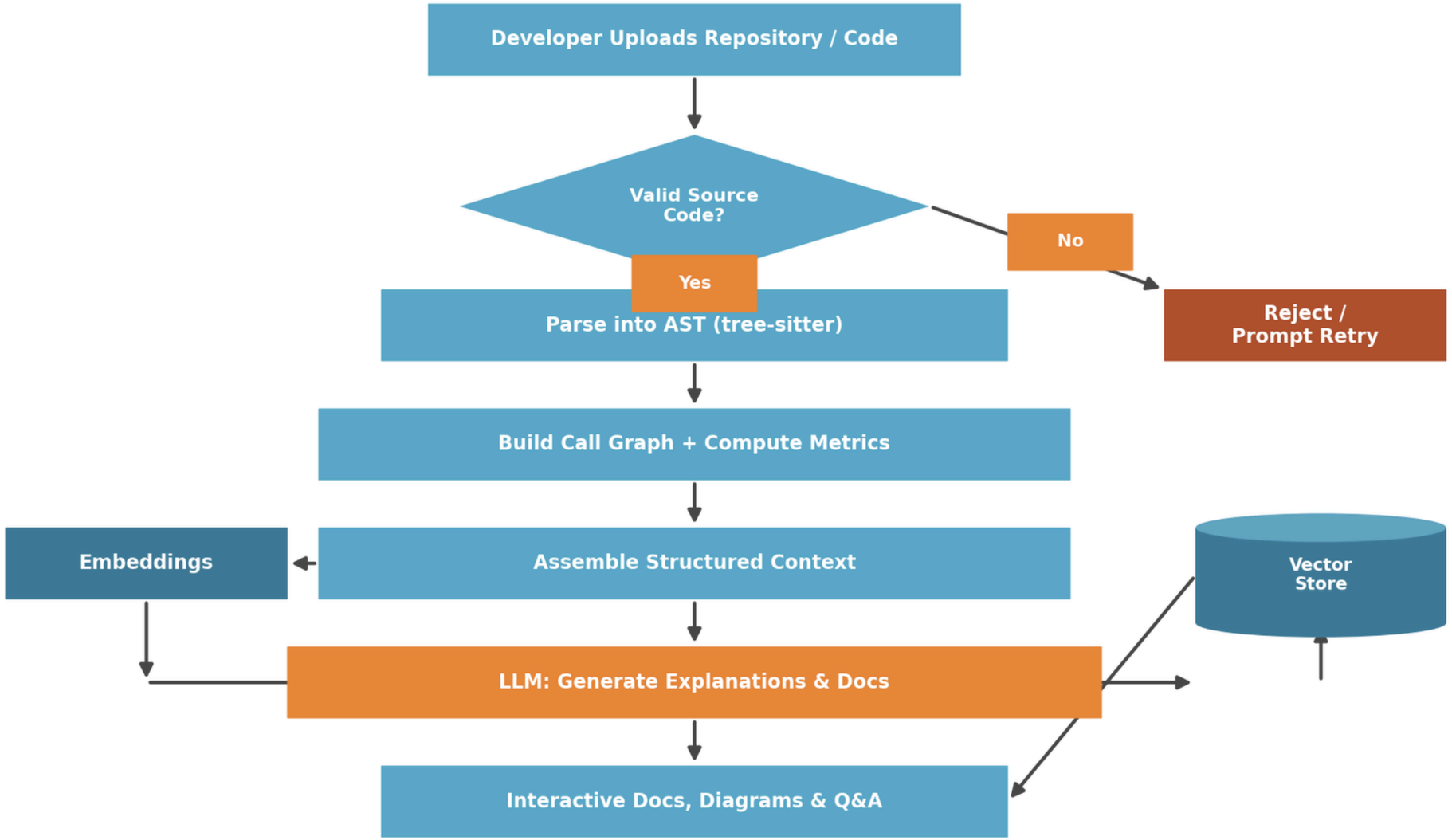
Project Objectives

- **Automated Code Comprehension:** Generate clear, plain-English explanations of files, classes, and functions in an unfamiliar codebase.
- **Architecture Visualization:** Construct and display call graphs and module-dependency diagrams that reveal how the system is structured.
- **Automated Documentation:** Produce the module- and function-level documentation that projects usually lack, directly from the source code.
- **Code-Quality Analysis:** Compute metrics such as cyclomatic complexity and detect dead code to highlight maintenance concerns.
- **Natural-Language Code Search:** Let developers ask questions about the codebase and retrieve relevant code by meaning using semantic search.
- **Grounded, Trustworthy AI:** Feed the LLM structured static-analysis output rather than raw code, reducing hallucination and improving reliability.

System Architecture



Data Flow Diagram



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References

[1] Z. Feng et al., “CodeBERT: A Pre-Trained Model for Programming and Natural Languages,” Findings of EMNLP, pp. 1536–1547, 2020.
[2] M. Chen et al., “Evaluating Large Language Models Trained on Code,” arXiv preprint arXiv:2107.03374, 2021.
[3] D. Guo et al., “GraphCodeBERT: Pre-training Code Representations with Data Flow,” Int. Conf. on Learning Representations (ICLR), 2021.
[4] X. Hu, G. Li, X. Xia, D. Lo, and Z. Jin, “Deep Code Comment Generation,” IEEE/ACM Int. Conf. on Program Comprehension (ICPC), pp. 200–210, 2018.
[5] W. U. Ahmad, S. Chakraborty, B. Ray, and K. Chang, “A Transformer-based Approach for Source Code Summarization,” ACL, pp. 4998–5007, 2020.
[6] M. Allamanis, E. T. Barr, P. Devanbu, and C. Sutton, “A Survey of Machine Learning for Big Code and Naturalness,” ACM Computing Surveys, vol. 51, no. 4, pp. 1–37, 2018.

Presentation





